



AUTHOR HEXSEC

FIREWALL SECURITY COMMANDS

By HexSec

FIREWALL

Display the firewall rules table

```
pfctl -sr
```

Show the NAT rules table

```
pfctl -sn
```

Reload the firewall rules

```
pfctl -f /etc/pf.conf
```

Reload filters and maintain state

```
pfctl -d -f /etc/pf.conf
```

Enable SSH with auto-generated rules

```
pfSsh.php playback enableallowallwan
```

Block an IP (adds rule to block list)

```
pfctl -t pfB_Blocklist -T add 203.0.113.50
```



hex_sec



hexsec_tools



hexsecteam

CHECK POINT

```
# List all firewall rules
# show access-rulebase all

# Add rule to allow SSH from any to local
# add access-rule src any dst 0
  action accept position 1

# Allow outgoing HTTP, HTTPS for 'internal'
  zone
# set access-rule from internal to any
  service http action accept

# Remove the first rule from rulebase
# delete access-rule 1

# Show all active connections
# fw tab -t connections -s

# Block an IP address temporarily
# block 192.168.1.100
```



hex_sec



hexsec_tools



hexsecteam

CISCO ASA

Enter configuration mode

enable

configure terminal

Show first 10 rules in the ACL

show access-list | begin access-lst | head 10

Allow and then remove SSH access

access-list 100 permit tcp any any any 22

no access-list 100 permit tcp any any eq 22

Allow traffic from a subnet

access-list 100 permit ip 192.168.1.0

255.255.255.0 any

Create and apply an ACL on an interface

access-list 101 deny icmp any any

access-group 101 in interface outside

Save the configuration

write memory



hex_sec



hexsec_tools



hexsecteam

FIREWALL

```
# List all firewall rules
```

```
show firewall
```

```
# Allow SSH to 192.168.1.50
```

```
set firewall name LAN rule 10 action 'accept'
```

```
set firewall name LAN rule 10 source address
```

```
192.168.1.0/24 set firewall name LAN rule 10
```

```
destination address 192.168.1.50
```

```
set firewall name LAN rule 10 destin port 'ssh'
```

```
# Deny all traffic from 203.0.113.0/24
```

```
set firewall name LAN rule 20 action 'drop'
```

```
set firewall name LAN rule 20 source address
```

```
203.0.113.0/24
```

```
# Enable connection tracking
```

```
set firewall conntrack enable
```

```
# Show firewall stats and counters
```

```
show firewall statistics
```



hex_sec



hexsec_tools



hexsecteam

FIREWALL

```
# Set default policy to drop traffic
set security policies default-policy drop

# Enable interface ge-0/0/1 in VLAN 100
set interfaces ge-0/0/1 unit 0
family inet vlan-id 100

# Allow SSH to the firewall
set security zones security-zone mgmt
interfaces ge-0/0/1 host-inbound-traffic
system services ssh

# Allow incoming HTTP traffic
set security policies from-zone untrust
to-zone trust policy allow-http
match source-address any destination-ad-
dress any application any
then permit

# Remove a policy by name
delete security policies policy allow-http

# List all (non-default) policies
show security policies
```

CLOUDFLARE

List existing firewall rules

```
cloudflare firewall rules list
```

Create & enable rule to block an IP

```
cloudflare firewall rules create --action  
block --filter "(ip.src eq 192.168.1.100)"  
--enabled true
```

Allow only country Canada

```
cloudflare firewall rules create --action  
allow --filter "(ip.geoip.country eq "CA")"
```

Block requests containing SQLi

```
cloudflare firewall rules create --action  
block --filter "(http.request.uri.query  
contains "UNION SELECT")"
```

Allow requests to admin panel

```
cloudflare firewall rules create --action  
allow --filter "(http.request.uri.path  
eq "/admin")"
```



hex_sec



hexsec_tools



hexsecteam

PALO ALTO

```
# Set management IP and default gateway
sudo deviceconfig system ip-address 192.168.0.1
netmask 255.255.255.0 default-gateway 192.168
0.254

# Create address, service and interface objects
set address Web-Server ip-netmask 10.0.0.10/32
set service TCP-HTTPS protocol TCP port 443
set network interface ethernet ethernet1/1
layer3

# Assign interface to a zone
set zone name DMZ network layer
ethernet1/1

# Create a security policy (allow https traffic)
set rulebase security rules Allow-HTTPS
from Trust to DMZ service TCP-HTTPS action
allow

# Commit changes
commit
```



hex_sec



hexsec_tools



hexsecteam

AZURE

```
# List all network security groups
az network nsg list -o table
```

```
# Create a new network security group
az network nsg create -g MyResourceGroup -n
MyNSG
```

```
# Allow inbound traffic on port 22 from any IP
az network nsg rule create -g MyResourceGroup
--nsg-name MyNSG -n AllowSSH --priority 1000
--destination-port-ranges 22 -access Allow
```

```
# Deny traffic to destination port 25
az network nsg rule create -g MyResourceGroup
--nsg-name MyNSG -n DenySMTP --priority 200
--destination-port-ranges 25 -access Deny
```

```
# List security rules for a network security
group
az network nsg rule list -g MyResourceGroup
--nsg-name MyNSG -o table
```

```
# Delete a security rule by name
az network nsg rule delete -g MyResureGroup
--nsg-name MyNSG -n DenySMTP
```


FIREWALL

```
# Enable or disable the firewall  
enable/disable
```

```
# Show firewall rules  
show firewall
```

```
# Add 'allow' rule, then commit  
changes and save  
set firewall rule new 20.1.1.100/32  
any any any  
firewall commit  
write memory
```

```
# Add 'deny' rule (no commit/save)  
set firewall rule new 192.168.0.0/  
any any deny
```

```
# Delete a firewall rule with ID 2  
set firewall delete 2
```

```
# Show dropped packets  
show firewall-drops
```

FIREWALL

```
# Enable the firewall
/ip firewall enable 0

# Show all firewall rules
/ip firewall rule print

# Allow SSH and Winbox access
/ip firewall filter add chain=input
    protocol=tcp dst-port=22 action=accept

# Allow established and related connections
/ip firewall filter add chain=input
    connection-state=established,related
    action=accept

# Drop invalid packets
/ip firewall filter add chain=input
    connection-state=invalid action=drop

# Block connections from an IP address
/ip firewall address-list add
    list=blocked address=192.168.1.100

# Log and drop packets from blocked IPs
/ip firewall filter add chain=input
    src-address-list=blocked action=drop
```

FIREWALLD

Enable and start firewalld

```
sudo systemctl enable --now firewalld
```

List default and active firewalld zone

```
sudo firewall-cmd --get-active-zones
```

Allow ports 80 and 443 (with reload)

```
sudo firewall-cmd --add-port=80/tcp  
--add-port=443/tcp --permanent --reload
```

Allow HTTP service (without reloading)

```
sudo firewall-cmd --add-service=http  
--permanent
```

Remove a service from the public zone

```
sudo firewall-cmd --remove-service=ssh  
--zone=public --permanent
```

Block an IP (rich rule example)

```
sudo firewall-cmd --add-rich-rule="rule  
family=ipv4 source address=192.168.1.100
```

FIREWALL

Enter configuration mode

configure terminal

Show existing access rules

show access-rules

Allow HTTP traffic from the LAN to WAN

access-rule from LAN to WAN service HTTP

action allow

Deny SSH from admin to all zones

access-rule from admin to any service

SSH action deny

Delete an access rule (by index number)

no access-rule 1

Enable the firewall

no sh

Block an IP via the CLI wizard

wizard block-side <IP>



[hex_sec](#)



[hexsec_tools](#)



[hexsecteam](#)

AWS

List all security groups

```
aws ec2 describe-security-groups
```

Create a new security group

```
aws ec2 create-security-group -group-name  
    MyGroup \  
    --description "my group"
```

Allow inbound traffic on port 22 (SSH)

```
aws ec2 authorize-security-group-ingress  
    --group-name MyGroup \  
    --protocol tcp --port 22 -cidr 0.0.0/0
```

Revoke inbound traffic on port 80 (HTTP)

```
aws ec2 revoke-security-group-ingress  
    --group-id sg-abcd1234 \  
    --protocol tcp --port 80 -cidr 0.0.0/0
```

Add an outbound rule (HTTPS on all IPs)

```
aws ec2 authorize-security-group-egress  
    --group-id sg-abcd1234 \  
    --protocol tcp --port 443 -cidr 0.0.0/0
```

Delete a security group

```
aws ec2 delete-security-group
```


FIREWALL

```
# Enable Windows Firewall
Set-NetFirewallProfile -Enabled True

# List all firewall rules
Get-NetFirewallRule

# Allow inbound traffic on ports 80 and 443
New-NetFirewallRule -DisplayName "Allow80+43"
-Direction Inbound -LocalPort 80,443
-Protocol TCP -Action Allow

# Create an outbound rule to block ICMP
New-NetFirewallRule -DisplayName "BlockICMP"
-Direction Outbound -Protocol ICMP
-Action Block

# Delete a firewall rule named "FTP"
Remove-NetFirewallRule -DisplayName "FTP"

# Show all allowed inbound rules
Get-NetFirewallRule -Direction Inbound
| Where-Object Action -EQ "Allow"
```

IPFIRE

```
# Enable IPFire on boot  
sudo ipfire start enable
```

```
# List all firewall rules  
sudo ipfire list
```

```
# Allow SSH access from a IP address  
sudo ipfire add allow ssh from 192.1.8.100
```

```
# Block incoming traffic to port 80  
sudo ipfire add deny all to 0/0 port 80
```

```
# Delete rule with ID 1000  
sudo ipfire delete 1000
```

```
# Reload firewall rules & reset state  
sudo ipfire drop
```

```
# Log and drop incoming packets  
sudo ipfire add log drop
```

```
# Toggle log settings set logging=on  
sudo ipfire log 200
```

NFTABLES

List all rulesets

> sudo nft list ruleset

Load rules from a file

> sudo nft -f /etc/nftables.conf

Allow incoming TCP ports 22 & 443

> sudo nft add rule inet filter input
tcp dport {22, 443} accept

Drop incoming traffic from an IP

> sudo nft add rule inet filter input
ip saddr 198.51.100.10 drop

Enable masquerading on eth0 (IPv4)

> sudo nft add rule inet nat postrouting
oifname eth0 masquerade

Set default policy to drop

> sudo nft chain inet filter input
{ policy drop; }

OPNSENSE

List firewall rules in the console

`pfctl -sr`

Reload all active firewall rules

`pfctl -f /tmp/rules.debug`

Allow outgoing HTTP traffic

`pfctl -a guest -sr lawk '!block &&/http/'`

Block Google DNS IP (8.8.8.8)

`pfctl -t TableName -T add 8.8.8.8`

Show entries in the table 'TableName'

`pfctl -t TableName -T show`

Delete a table definition

`pfctl -T kill TableName`

Enable logging for the 'lan' interface

`ifconfig lan0 setinfo flow`

Disable firewall (stop packet filter)

`pfctl -d`